



Total Parenteral Nutrition Procedure

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	All clinical services	Nursing, Medical, Dietetics and Pharmacists

2. Purpose

This procedure describes the safe and effective provision of a total parenteral nutrition (TPN) service within the Armada Kalamunda Group, the standards required, the outcomes to be monitored and staff responsibilities and accountabilities.

- TPN refers to the provision of nutrition via a central vein and is required when oral and enteral nutrition is insufficient or unsafe.
- TPN is continuously infused over 24 hours via a controlled infusion pump into a central high flow vein.
- It may be administered via a central venous line (CVC) or via a peripherally inserted central catheter (PICC).
- Some patients receive overnight TPN infusions, and long-term parenteral nutrition can be delivered at home safely.
- The TPN formulation is determined by the patient's clinical condition, but the nutritional requirements of most patients can usually be adequately supplied using standard formulations.
- The prescription of TPN is the result of collaboration between the patient's health care team including the treating medical officer (MO), dietitian, ICU consultant / senior MO, nurses and pharmacist.

3. Consideration

3.1 Indications for TPN

- Inability to absorb nutrients via the gastrointestinal (GI) tract,
 - Short bowel syndrome.
 - Malabsorption.
 - Obstruction.
- Severe catabolism with a temporary (>three days) non-functional GI tract.
- Major surgery when adequate enteral intake is not expected to resume within five-seven days.
- Entero-cutaneous or high-output fistulae.
- Inflammatory bowel disease when temporary bowel rest is indicated.
- Prolonged ileus (>three days).

3.2 Contraindications for TPN

- Nutritional requirements can be met either via oral or enteral nutrition.
- The patient is end of life care (palliative).
- Well-nourished patient, expected to return to normal oral / enteral nutrition in a five - seven days.

3.3 Home TPN

- Patients requiring long term TPN may be discharged home with self-managed parenteral nutrition.
 - AKG does not initiate home TPN, these patients should be managed from a tertiary centre with experience in this service.

4. Roles and responsibilities

Role	Responsibility
Medical Officer	• Ensure daily morning blood samples are ordered • Replacement of deficient electrolytes prior to commencement of TPN. • Review maintenance fluids • Review daily biochemistry results. • Ensure that central venous access is present and patent. • Monitor weight, fluid balance, biochemistry, blood sugar levels, aspirates and bowel function or stoma output.
Dietitian	• Complete nutrition assessment: anthropometry, biochemistry, clinical and diet history. • Establish re-feeding risk. • Calculate energy and protein requirements (including the ratio of glucose, lipid and amino acids). • Provide recommendation on starting TPN formula and rates of escalation to final formula. • Transition from PN to oral or enteral nutrition. • Advises on monitoring requirements for patients at risk of vitamin, mineral and trace element abnormalities.
Nurse	• Oversee the care of the vascular access device (VAD). • Manage the infusion and associated equipment. • Ensure accurate patient weight and height is documented at admission and then weekly or as required by the team. • Perform and document regular blood glucose levels (BGL). • Maintain accurate fluid balance chart. Liaise with the MO to incorporate maintenance IV fluid requirements into TPN prescription as necessary.

Role	Responsibility
	Monitor patient and notify MO of improvements or deterioration in GI function including symptoms of nausea and/or vomiting, diarrhoea/constipation, abdominal distension.
Pharmacist	- Liaise regarding stock procurement. - Ensures all medication are given by the most appropriate route of administration taking into account patient's ability to absorb medications enterally. - Review potential interactions between feed and patient medication regimen.

5. Procedure

5.1 Ordering TPN

- Referral to the TPN team, contact the ICU Consultant / Registrar ext. 2784 and/or ICU dietitian page 226 prior to commencement.
- Prior to commencing TPN, the following tests should be conducted:
 - Electrolytes (Na, K, PO₄³⁻, Mg²⁺).
 - Urea, creatinine.
 - LFTs.
 - Cholesterol / Triglycerides (TG).
 - Note: Do not measure TG while TPN is running as this gives unreliable results.
 - INR, Prothrombin Time.
 - Magnesium.
 - Full blood count.
 - Vitamins / trace elements if malnourished (e.g. Zn, Fe, Vit A, D, E, B12).
- Initial nutritional assessment is conducted by the consultant / registrar, dietitian, and nurse.
- The MO and dietitian calculate nutritional requirements; refer to [Appendix 1 Nutrient Considerations](#) for additional information regarding nutrient requirements.
- TPN is to be prescribed by the MO on the Total Parenteral Nutrition Order Form AKMR 176.3.
- The Total Parenteral Nutrition Order Form must have the signature, name and designations of the medical officer and dietitian.
- The Total Parenteral Nutrition Order Form initiation and rate changes are to be signed off by the ICU consultant/registrar on initiation of TPN until target rate is reached and parameters are stable. Once patient is stable Total Parenteral Nutrition Order Form can be signed off by treating medical team.
- The nurse sends the TPN order to pharmacy.
- TPN solution is commercially sourced by pharmacy; refer to [Appendix 2 Armadale Standard Formula](#).
- The TPN order is to be sent to pharmacy by 10:30am.

5.2 Prescribing TPN

- TPN should be introduced progressively, usually starting at no more than 50% of estimated needs for the first 24-48 hours.
- Continuous administration of TPN is the preferred method of infusion.
 - The usual starting rate is approximately 40 mL per hour increasing by 30mL per hour every 24 hours to 85-90mL per hour.
- The total volume of TPN mixture is infused at an hourly rate over a 24-hour period.
 - The hourly rate of TPN solution is calculated by dividing the total volume of TPN base solution by 24 hours.
- A change from continuous to cyclical / overnight delivery should be considered for patients requiring TPN for more than two weeks or on home TPN, refer to 5.7 Cyclical / Overnight TPN.
- If potassium, magnesium and / or phosphate are low, give supplements intravenously before starting TPN.
- TPN should be started slowly and only increased when the patient exhibits tolerance to the glucose load and the electrolyte deficiencies have been corrected.
- For patients not at risk of refeeding syndrome aim to commence at 50% of goal rate for first 24 hours.
 - If tolerating TPN, increase to the goal rate as instructed by TPN Team.
- Refeeding Syndrome: For patients at risk of refeeding syndrome give a maximum of 42 kJ / kg / day (10 kcal / kg / day) for first 24 hours.
- If TPN tolerated and electrolytes are monitored and corrected, gradually increase to target rate as directed by the TPN team.
- Daily vitamins and minerals should be components of all TPN prescriptions, refer to [Appendix 3 Micronutrient Preparations](#).

5.3 Administering TPN

- Two nurses, one of whom must be a registered nurse should perform the checking procedure.
- Once the clinical pharmacist has signed the form, the prescription **may not** be altered without consultation with the clinical pharmacist.
- Strict aseptic technique is required for the connection of any TPN infusion line to prevent contamination of the connection points.
 - Blood samples should not be drawn from the TPN line.
- All TPN should be commenced and changed at 1800 hours unless bag runs out.
- Program volumetric infusion pumps with a second person and commence infusion.
 - An inline 1.2 micron filter should be used for all fat-containing TPN preparations.
 - The solution should be sealed in the light protective over-pouches that are supplied with each bag as it is sensitive to light.
 - Light protective over pouch must be taped at the bottom during administration.
 - Hang time should be 24 hours.
 - Once activated the bag can be refrigerated with an expiry of seven days.
 - Label the bag clearly with the activation date and time.

- Where there is risk of re-feeding syndrome or metabolic complications, increase slowly as per medical order to patients target rate.
- Approval for co-infusion of insulin via the TPN line is to be documented in the notes by the MO, Clinical Pharmacist and / or Shift Coordinator.
 - Other medications should not be co-infused.
- Connect co-infusion at the time of the TPN infusion change.
 - If co-infusion is completed prior to completion of TPN, do not disconnect infusion lines until TPN has completed and the administration set needs to be changed.
- Administration sets should be changed every 24 hours.

5.4 TPN-specific monitoring standards

- Monitor overall clinical condition, biochemical stability, renal function, liver function, oedema, fluid balance and oral / enteral intake.
- The central venous access site is monitored daily.
- Check temperature and other vital signs four hourly for the duration of TPN therapy.
- Weigh the patient prior to commencing TPN, then at a regular time daily.
- Blood glucose levels (BGL) should be monitored closely e.g., two-six hourly:
 - on initiation of TPN.
 - on commencement of insulin, if required.
 - when TPN infusion or insulin infusion rate is changed.
 - on cessation of TPN.
- Once stable, monitor BGL as clinically indicated.
 - The target BGL range is to be set by the treating MO.
- Observations must be recorded from commencement of TPN until TPN is ceased.

Table 1: Recommended observations

Parameters	Rationale for monitoring	Frequency of monitoring
Routine observations: Temperature Pulse rate Respiratory rate Blood pressure	Infection, cardiac function, fluid status, response to feeding	Four hourly, then eight hourly when stable.
Fluid balance	Fluid status	Hourly, totalled twice daily.
Body weight	Fluid status	Daily until stable, then weekly.
Blood glucose levels	Glucose tolerance	Six hourly, together with insulin sliding scale if required. Less when stable
Na, K, urea, creatinine	Renal function, response to refeeding	Daily, then three times a week when stable.
Ca, Albumin	Response to refeeding	Daily, then twice weekly when stable.

Liver function tests	Liver function	Daily, then three times a week when stable.
Full Blood Count, prothrombin time	Haemodynamic stability	Daily, then twice weekly when stable.
Phosphate, Magnesium	Response to refeeding	Daily, then three times a week when stable.
Serum triglyceride, cholesterol	Lipid tolerance	Weekly until stable, then three monthly.
Iron	Long term status	Baseline, then three monthly

5.5 TPN documentation standards

- Record commencement of TPN in the patient health record, and document TPN therapy on Nursing Care Plan.
- Document the time the infusion was commenced on the TPN order form.
 - Both nurses checking the TPN should sign the form.
- Rate changes must be checked, dated, time recorded and signed by two staff members, one of whom must be a registered nurse, on the fluid balance chart and in the patient health record.
- Document TPN volumes infused hourly on fluid balance chart or ICU Observation chart, AKMR 147C.
- Maintain an accurate fluid balance chart.
- Maintain a general observation chart.
- Maintain a diabetic record chart or an Insulin medication chart as appropriate.
- Record any problems associated with TPN administration and actions taken e.g., line breakage, accidental disconnection, filter breakages, central VAD compromise.

5.6 Ceasing TPN

- TPN can be withdrawn once adequate oral or enteral nutrition is tolerated and nutritional status is stable.
- Liaise with the dietitian if transitioning to enteral nutrition.
- Abrupt cessation of TPN should be avoided where possible to reduce the risk of hypoglycaemia.
- If TPN is abruptly ceased, the patient should receive IV dextrose / insulin infusion.
- Care should be taken when ceasing TPN on patients receiving insulin.
 - Cease the insulin before reducing TPN to avoid hypoglycaemia.
 - Time delay may depend on the type of insulin and oral intake.
- Monitor blood glucose levels every hour for the first two-four hours, then every two-six hours afterwards until stable.
- When tapering, reduce the TPN rate by 30mL per hour every two to four hours.
 - Liaise with the MO regarding rate reduction protocol.
- Rate changes to be prescribed on the TPN order form.

5.7 Cyclical / Overnight TPN

- Used for some inpatients or long-term home TPN patients.
- Generally administered over 12-16 hours.
- Rates are determined by dietitian and MO.
- MO is to prescribe TPN infusion on the TPN request form.
- Usually, TPN is started in the evening at a low rate.
- Increase in two-four steps to a maximum rate, and then infuse at the maximum rate for a designated number of hours.
- Reduce the rate in steps (the reverse of the increasing regimen) before ceasing.

5.8 Troubleshooting TPN

- If in doubt, contact the ICU registrar
- If TPN runs out, obtain order from MO for a 10% dextrose solution to be administered at the same rate as TPN was running.
- If CVL or PICC line is not functional, contact ICU registrar and administer 10% dextrose as above.
 - **Do not** re-connect a disconnected TPN line.
- If the TPN bag does not correspond to the TPN prescription, do not connect infusion, and contact the pharmacist or the ICU registrar.

5.9 Storage

- The multi-chamber bags whilst remaining in the outer wrapper and protected from light can be stored for up to 12 months from the date of manufacture at room temperature (less than 25 degrees Celsius).
- The shelf life is altered only if exposed to light or if removed from the outer plastic wrapper containing the oxygen absorber.
 - Once taken out of the plastic wrapper the shelf life is reduced to seven days, including hanging.

6. Legislative context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) [Clinical Incident Management Policy MP 0122/19](#)
- HDWA [Insertion and Management of Peripheral Intravenous Cannulae in Western Australian Healthcare Facilities MP 0038/15](#)

7. Supporting information

The following documents support the implementation of this procedure:

- [Central Venous Catheters Insertion and Management Procedure GEN-PRO-0076-17](#)
- [Peripherally Inserted Central Catheter Care of Procedure AKG-PRO-0354-17](#)
- [Aseptic Technique Procedure AKG-PRO-0045-14](#)
- [Nutrition and Hydration Policy AKG-POL-0603/23](#)
- [RPMG, Clinical Practice Standard for Parenteral Nutrition: Total \(TPN\) Management, 2023](#)

8. Compliance, monitoring and evaluation

Compliance against this policy will be evaluated with routine incident review processes and safety and quality reporting mechanisms in the ICU and surgical services including reviewing of related clinical incidents as recorded in Datix CIMS.

9. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.3 Partnering with consumers
- Clinicians use organisational processes from the Partnering with Consumers Standard when preventing and managing healthcare-associated infections, and implementing the antimicrobial stewardship program to:
 - a. Actively involve patients in their own care
 - b. Meet the patient's information needs
 - c. Share decision-making
- Standard 3 – Action 3.9 Aseptic Technique
- The health service organisation has processes for aseptic technique that:
 - a. Identify the procedures where aseptic technique applies
 - b. Assess the competence of the workforce in performing aseptic technique
 - c. Provide training to address gaps in competency
 - d. Monitor compliance with the organisation's policies on aseptic technique
- Standard 3 – Action 3.10 Invasive medical devices
- The health service organisation has processes for the appropriate use and management of invasive medical devices that are consistent with the current edition of the Australian Guidelines for the Prevention and Control of Infection in Healthcare¹¹²
- Standard 5 – Action 5.28
- The workforce uses the systems for preparation and distribution of food and fluids to:
 - a. Meet patients' nutritional needs and requirements
 - b. Monitor the nutritional care of patients at risk
 - c. Identify, and provide access to, nutritional support for patients who cannot meet their nutritional requirements with food alone
 - d. Support patients who require assistance with eating and drinking

10. References

Acknowledgement of other site endorsed guidelines used to compile this guideline.

1. SMHS, Total Parenteral Nutrition (TPN) Clinical Practice Standard, 2013
2. State of Queensland (Queensland Health) Parenteral Nutrition Administration and Management Guidelines, 2011
3. Parenteral Nutrition Working Party, Toowoomba Hospital, Darling Downs and West Moreton Health Service District

11. Policy contact (relating to these procedures)

Enquiries relating to this procedure may be directed to:

Title: Coordinator Dietetics and Nutrition
Department: Dietetics
Email: jane.priest@health.wa.gov.au

12. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
Version 3	07/09/2023	07/09/2026	Scheduled review – minor amendment

13. Authorisation

This procedure has been authorised and issued by the Director Allied Health.

Authorised by	A/Director Allied Health
Authorisation date	My Linh Nguyen
Published date	07/09/2023

Appendix 1: Nutrient Considerations

- Determination of the nutrient requirements should be individualised, based on assessment of body composition and function, and fall within acceptable ranges, while taking physiological and pathophysiological conditions into account.
 - Care should be taken not to overfeed or underfeed the patient.
- To minimise the risk of overfeeding the dietitian will use actual body weight if the patient is underweight or healthy weight or an adjusted body weight if Body Mass Index (BMI) $>30\text{ kg / m}^2$.
- Adjusted ideal body weight (IBW) = $[(\text{Current weight} - \text{IBW}) \times 0.25] + \text{IBW}$.
- IBW weight at BMI = 25.

Carbohydrate: the minimum glucose required is 2 g / kg / day to meet the needs of the cells in the brain, kidney, and erythrocytes. The maximum infusion rate of carbohydrate should be no more than 4 mg / kg / min or 5-10 g / kg / day.

- This reflects the estimated maximum limit for glucose oxidation.
- Problems with excessive carbohydrate include hyperglycaemia, hepatic steatosis, excessive CO₂ production, predisposition to sepsis and delayed gastric emptying.

Fat: The maximum amount of fat should be no more than 2 g / kg / day.

Protein (Amino Acids): Protein requirements are estimated at 1.0-1.5 g / kg / day and 1.5-2.0 g / kg / day for critically ill patients.

- Exceeding 1.9 g protein / kg / day does not confer benefit and may be dangerous.
- Overfeeding protein to critically ill and injured trauma patients can lead to uraemia.

Fluid: 35-45 ml / kg / day or 0.24 ml / kJ / day (1 ml / kCal), plus extra fluid to accommodate for extra fluid losses.

- For obese patients, use 0.24 ml / kJ.

Appendix 2: Armadale standard formula (Formula 1)

TPN Solution

- Multi chamber bags are easy, convenient, and safe to use and reduce the risk of contamination.
- Balanced nutrient and energy requirements can be met with a combination of amino acid, glucose solutions and lipid emulsions.
- The nutrient profiles are designed to meet the requirements of 80% of the patients.
- Customised formulations are to be discussed with the clinical pharmacist prior to ordering.
- There is a 24–72-hour turnaround time for the supply of customised bags as bags are provided by external supplier.
- Any customised orders must be received by the pharmacy by 10:30am.

Armadale standard formula breakdown (Formula 1)

Description	Expiry
OLIMEL N9-840E 1500ml Amino Acids with electrolytes 14.2% Glucose 27.5% ClinOleic 20%	Two years at room temperature.

Nutrient	Armadale Formula
Energy	6720kJ / 1600kcal
Non-protein Energy	5292kJ / 1260kcal
Amino Acids	85.4g
Nitrogen	13.5g
Glucose	165g
Lipid	60g
Sodium	52.5mmol
Potassium	45mmol
Calcium	5.3mmol
Magnesium	6mmol
Phosphate *	22.5mmol
Chloride	68mmol
Acetate	80mmol
Additions	Nil additions to standard formula

*Includes phosphate from lipid emulsion

Appendix 3: Micronutrient Preparations

- Vitamins and trace elements should be components of all TPN prescriptions.
- CERNEVIT is a multivitamin preparation, lyophilised, sterile powder, for reconstitution in 5 mL of water for injection or other compatible parenteral fluids.
- The micronutrient prescription should be documented on the medication chart as directed in *Table 1*.
- **Do not add anything into the TPN bag itself.**

Table 1: Recommended micronutrient administration schedule

Drug	Dose	Frequency and Time	Route	Administration
Cernevit Single dose of Cernevit is reconstituted by adding 5ml sterile water for injection or other intravenous fluids (0.9% sodium chloride injection).	One Vial (5ml)	Daily	Intravenous	Deliver combined in 100ml of 5% glucose or 0.9% normal saline over 60 minutes but not exceeding four hours.
Trace Elements Baxter ADTE	One syringe			

Table 2: ADTE Trace Element composition

Element	ADTE (per 10mL)
Zinc	100 µmol (6.5mg)
Copper	8 µmol (508µg)
Selenium	1 µmol (80µg)
Iron	20 µmol (1.1mg)
Manganese	1 µmol (55µg)
Chromium	0.2 µmol (10µg)
Molybdenum	0.2 µmol (19µg)
Iodine	1.0 µmol (130µg)

Table 3: Cernevit multivitamin composition

Active ingredients	Quantity	Corresponding to	Quantity
Retinol (present as retinyl palminate)	3500 IU	Vitamin A	3500 IU
Cholecalciferol	5.51 µg	Vitamin D3	220 µg
dl-α-tocopherol	10.20 mg	Alpha tocopherol (Vitamin E)	11.20 IU
Ascorbic acid	125 mg	Vitamin C	125 mg
Coccarboxylase tetrahydrate	5.80 mg	Thiamine (Vitamin B1)	3.51 mg
Riboflavine sodium phosphate	5.67 mg	Riboflavine (Vitamin B2)	4.14 mg
Pyridoxine hydrochloride	5.50 mg	Pyridoxine (Vitamin B6)	4.53 mg
Cyanocobalamin	6 µg	Vitamin B12	6 µg
Folic acid	414 µg	Folic acid (Vitamin B9)	414 µg
Dexpanthenol	16.15 mg	Dexpanthanoic acid (Vitamin B5)	17.25 mg
d-Biotin	69 µg	Biotin (Vitamin B7)	69 µg
Nicotinamide	46 mg	Niacin (Vitamin PP / B3)	46 mg